REVISION NOTES

Week 2

Mon 12 – Sun 18 October 2026

22	10 h 13 min	22 / 22
LECTURES	VIDEO	SUMMARISED

COURSES THIS WEEK

MATH 110 Linear Algebra	6 lectures · 1 h 52 min
AI 200 Neural Networks: Zero to Hero	3 lectures · 3 h 52 min
SPCH 100 Storytelling	5 lectures · 1 h 16 min
MATH 120 Single-Variable Calculus	6 lectures · 1 h 34 min
MATH 130 Probability	2 lectures · 1 h 39 min
Total	**22 lectures · 10 h 13 min**

Study days Mon 12 Oct to Sun 18 Oct; Sat 17 Oct is the rest day. Hours are video time; a lecture the plan spreads over several days counts only the days that fall in this week.

WEEK 2 · MON 12 – SUN 18 OCTOBER 2026

Contents

LECTURES, IN PLAN ORDER

1	MATH 110	Cramer's rule, explained geometrically	Mon 12
2	MATH 110	Change of basis	Mon 12
3	MATH 110	Eigenvectors and eigenvalues	Mon 12
4	MATH 110	A quick trick for computing eigenvalues	Mon 12
5	AI 200	Makemore 1 — bigram character-level language model	Mon 12 – Tue 13
6	SPCH 100	What is a story — and why tell one	Mon 12
7	MATH 120	Limits, L'Hôpital's rule, and epsilon delta definitions	Tue 13
8	MATH 120	Integration and the fundamental theorem of calculus	Tue 13
9	MATH 120	What does area have to do with slope?	Tue 13
10	MATH 120	Higher order derivatives	Tue 13
11	SPCH 100	The five-second moment: the instant a story is about	Tue 13 – Wed 14
12	MATH 130	Birthday Problem, Properties of Probability	Wed 14
13	AI 200	Makemore 2 — MLP, following Bengio et al. 2003	Wed 14 – Thu 15
14	MATH 110	Abstract vector spaces	Thu 15
15	MATH 110	The Geometry of Linear Equations	Thu 15
16	SPCH 100	Anecdote, bait and reflection — the two building blocks	Thu 15
17	SPCH 100	The elephant: a reason to listen in the first thirty seconds	Thu 15
18	MATH 120	Taylor series	Fri 16
19	MATH 120	The other way to visualize derivatives	Fri 16
20	AI 200	Makemore 3 — activations, gradients, BatchNorm	Fri 16 – Sun 18
21	MATH 130	Conditional Probability	Sun 18
22	SPCH 100	The clues to a great story — make me care	Sun 18

AT THE BACK

What this week established

Answers to the self-checks

Every lecture this week has a summary.

MATH 110 · LINEAR ALGEBRA · LECTURE 1 OF 22

Cramer's rule, explained geometrically

Mon 12 Oct · 12 min video · Watch on YouTube

THE TAKEAWAY

Each coordinate of the unknown vector is an area, and every area scales by the determinant — that is the whole of Cramer's rule.

THE ARGUMENT · 5 STEPS

1. **A wrong idea, pointing the right way**
   You might hope that dotting the output with the transformed basis vectors recovers the coordinates. It does not: transformations rarely preserve dot products. The ones that do are called orthonormal, and for those the hope is exactly right.

2. **A coordinate, written as an area**
   The parallelogram spanned by $\hat{\imath}$ and the unknown vector has base 1, so its signed area IS that vector's y-coordinate. Pairing the unknown vector with $\hat{\jmath}$ gives the x-coordinate the same way.

3. **In three dimensions it is a volume**
   Pair the unknown vector with the two basis vectors you are not asking about. The signed volume of that parallelepiped is the coordinate you want, and the right-hand rule keeps the sign honest.

4. **Areas do not survive, but they all scale alike**
   That is what the determinant is. After the transformation every parallelogram's area has been multiplied by $\det A$ — the same factor for all of them — so dividing by it undoes the damage.

5. **Which makes it computable from the output alone**
   Replace one column of A with the output vector and take that determinant, then divide by $\det A$. Replacing the first column gives x, the second gives y. Nothing but numbers you were handed.

WORKED EXAMPLE

$A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$, $\vec{v} = (9, 8)$. Swap the output into column one: $\det \begin{pmatrix} 9 & 1 \\ 8 & 2 \end{pmatrix} = 10$, and $\det A = 5$. So $x = 10/5 = 2$, and back-substituting gives $y = 3$.

WATCH FOR

Reaching for Cramer's rule to actually solve systems. Gaussian elimination is always faster; this exists to show the determinant and the solution are one fact.

KEY CONCEPTS

The determinant is a volume factor

CHECK YOURSELF · ANSWERS AT THE BACK

1.1 Cramer's rule works because a linear transformation:

A Preserves all areas
B Scales every area by the same factor, its determinant
C Preserves dot products
D Keeps the basis vectors perpendicular

1.2 For $A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ and $\vec{v} = (9, 8)$, Cramer's rule gives $x =$

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 2 OF 22

Change of basis

Mon 12 Oct · 13 min video · Watch on YouTube

THE TAKEAWAY

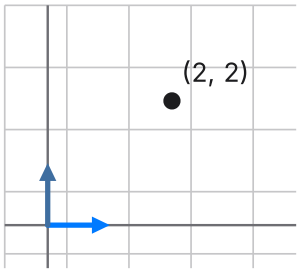
A matrix whose columns are someone else's basis vectors translates their coordinates into yours; its inverse goes the other way.

THE ARGUMENT · 5 STEPS

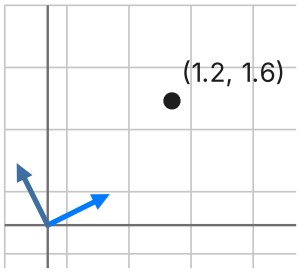
1. **Coordinates are a language, not the vector**
   $(3, 2)$ means scale $\hat{\imath}$ by 3, scale $\hat{\jmath}$ by 2, add. Those two vectors carry every implicit assumption: which way is right, which is up, how long one unit is.

2. **Another basis is another language**
   Jennifer's $\vec{b}_1$ and $\vec{b}_2$ are $(2, 1)$ and $(-1, 1)$ in our coordinates. In hers they are $(1, 0)$ and $(0, 1)$ — they are what those coordinates MEAN in her world. Same arrows in space, different words.

3. **The grid is a construct**
   Space has no grid of its own; a grid is a drawing of a chosen basis. Only the origin is agreed on, because everybody's $(0, 0)$ is what you get by scaling anything by zero.



standard basis



another basis

4 **Her language into ours is a matrix-vector product**

Put her basis vectors, written in our coordinates, into the columns. Multiplying by her coordinates scales those columns and adds them — which is exactly what her coordinates were instructing all along.

5 **A transformation from her seat is $A^{-1}MA$**

Translate her vector into our language, apply the transformation there, translate the result back. The middle matrix is the transformation as we see it; the outer two are the change of perspective.

WORKED EXAMPLE

The vector she calls $(-1, 2)$: compute $-1 \cdot (2, 1) + 2 \cdot (-1, 1) = (-4, 1)$, and that is it in our coordinates. Going the other way, our $(3, 2)$ is her $(5/3,\ 1/3)$ — same arrow, inverse matrix.

WATCH FOR

Expecting the matrix to convert OUR coordinates into hers, because geometrically it moves our grid onto hers. Numerically it does the exact opposite.

KEY CONCEPTS

A basis is the ruler you measure with

CHECK YOURSELF · ANSWERS AT THE BACK

2.1 The expression $A^{-1}MA$ represents:

A The transformation M applied twice
B The same transformation M, described in the other basis
C The inverse of M
D A rotation composed with M

2.2 Her basis vectors are $(2, 1)$ and $(-1, 1)$ in our coordinates. The vector she calls $(-1, 2)$ has our first coordinate:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 3 OF 22

Eigenvectors and eigenvalues

Mon 12 Oct · 17 min video · Watch on YouTube

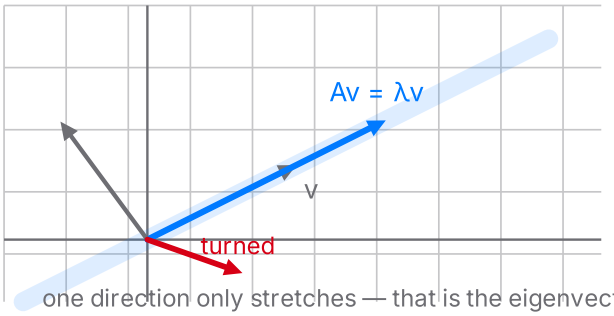
THE TAKEAWAY

An eigenvector is a direction the transformation only stretches, never turns, and its eigenvalue is that stretch.

THE ARGUMENT · 6 STEPS

1 **Most vectors get knocked off their line**

Apply a transformation and watch a vector: usually it ends up pointing somewhere new. Some special directions stay on their own line and are merely scaled. Those are the eigenvectors.



2 **The defining equation**

$Av = \lambda v$ with $v \neq 0$. The matrix on the left and the plain number on the right do the same thing to that one vector — which is why eigenvectors make matrix powers easy.

3 **Finding them means engineering a collapse**

Rewrite as $(A - \lambda I)v = 0$. A nonzero v can only be sent to zero if $A - \lambda I$ squashes space, so $\det(A - \lambda I) = 0$. That is the characteristic equation, and this is where the determinant earns its keep.

4 **Then solve for the direction**

Each root λ gives a matrix $A - \lambda I$ whose null space is the eigenvectors for that λ — a whole line of them, since scaling an eigenvector leaves it an eigenvector.

5 **Not every map has enough of them**

A shear has a single eigendirection; some rotations have none at all in the real plane, because every direction genuinely turns. An eigenbasis is a privilege, not a guarantee.

6 **Why anyone cares**

In an eigenbasis the transformation is just independent stretches along axes, so applying it a hundred times means raising numbers to the hundredth power instead of multiplying matrices.

WORKED EXAMPLE

$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$. Then $\det(A - \lambda I) = (3 - \lambda)(2 - \lambda)$, giving $\lambda = 3, 2$ — the diagonal, as always for a triangular matrix. For $\lambda = 3$, solve $(A - 3I)v = 0$: $\begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix} v = 0$ forces the second component to zero, so $v = (1, 0)$.

WATCH FOR

Assuming every matrix has a full set of eigenvectors. Ask what a shear does before you assume an eigenbasis exists.

KEY CONCEPTS

Eigenvectors are the directions that survive

CHECK YOURSELF · ANSWERS AT THE BACK

3.1 The characteristic equation is $\det(A - \lambda I) = 0$ because:

A Determinants are easy to compute
B A nonzero vector can only be sent to zero if the map collapses space
C Every matrix is invertible
D λ must be real

3.2 The largest eigenvalue of $\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$ is:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 4 OF 22

A quick trick for computing eigenvalues

Mon 12 Oct · 13 min video · Watch on YouTube

THE TAKEAWAY

The trace and the determinant hand you the mean and the product of the eigenvalues, and two numbers are pinned down by those.

THE ARGUMENT · 5 STEPS

1. **Fact one: the trace is the sum**
   The two diagonal entries add to the sum of the eigenvalues, so the mean of the eigenvalues is the mean of the diagonal — and you can read that off the matrix without writing anything down.

2. **Fact two: the determinant is the product**
   Which ought to feel right: an eigenvalue says how much one direction is stretched, and the determinant says how much area is scaled overall. Multiply the stretches and you have the area factor.

3. **Fact three: mean and product pin down the pair**
   Two numbers sitting at $m \pm d$ multiply to $m^2 - d^2$, a difference of squares. Set that equal to the product p and $d = \sqrt{m^2 - p}$ falls straight out.

4. **The whole trick in one line**
   $\lambda = m \pm \sqrt{m^2 - p}$, with m the mean of the diagonal and p the determinant. No characteristic polynomial, no expanding, no quadratic formula.

5 **Why it is faster, and when it is not**

It is the same mathematics as finding the roots of the characteristic polynomial — it has to be. The saving is that m and p are legible straight off the matrix, so you begin writing the answer immediately.

WORKED EXAMPLE

$\begin{pmatrix} 3 & 1 \\ 4 & 1 \end{pmatrix}$: the mean of 3 and 1 is 2, and the determinant is $3 - 4 = -1$. So the eigenvalues are $2 \pm \sqrt{4 + 1} = 2 \pm \sqrt{5}$. You never wrote down a polynomial.

WATCH FOR

Filing it as one more formula to memorise. Its value is that the trace and the determinant already mean something, so it carries its own derivation.

KEY CONCEPTS

Eigenvectors are the directions that survive

CHECK YOURSELF · ANSWERS AT THE BACK

4.1 In $\lambda = m \pm \sqrt{m^2 - p}$, the quantity p is:

A The trace
B The determinant
C The mean of the diagonal
D The larger eigenvalue

4.2 For $\begin{pmatrix} 2 & 7 \\ 1 & 8 \end{pmatrix}$, the larger eigenvalue is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 5 OF 22

Makemore 1 — bigram character-level language model

Mon 12 Oct – Tue 13 Oct · day 2–3 of 3 · 118 min video · Watch on YouTube · began in week 1

THE TAKEAWAY

A bigram language model can be built two ways — count which character follows which and normalise, or train a one-layer net with a softmax on the negative log likelihood — and the two arrive at the same model, which is the point: the neural framing is the one that scales.

THE ARGUMENT · 8 STEPS

1. **One file of names, many examples**
   32,033 names, 2 to 15 letters long. Each name is several examples at once: every character is evidence about which character follows it, and where names start and end.

2. **Counting bigrams**
   Wrap each name in start and end markers and count every adjacent pair. The final version uses one token, '.', at index 0 for both start and end, giving a 27 × 27 table N whose row i counts what follows character i.

3. **From counts to samples**
   Normalise each row into probabilities, start from '.', and draw the next character with torch.multinomial until '.' comes back. The names ('mor.', 'axx.', 'minaymoryles.') are bad — a bigram sees one character of context — but they are drawn from the data's statistics.

4. **Broadcasting, carefully**
   Normalise with P.sum(1, keepdims=True): a 27 × 1 column of row sums divides each ROW. P.sum(1) gives shape 27, which broadcasts as a 1 × 27 row and divides each COLUMN instead — no error, just a wrong model.

5. **Scoring a model: negative log likelihood**
   Maximising the likelihood of the data — the product of the probabilities assigned to every actual next character — is the same as minimising the average negative log likelihood. The smoothed count model scores 2.4765. Adding 1 to every count keeps any pair from having probability 0, which would make the loss infinite.

6. **The same model as a neural net**
   One-hot encode the input character and multiply by a 27 × 27 weight matrix W: that selects W's row for the character — 27 logits, read as log-counts. Exponentiate and normalise (together, a softmax) to get probabilities. The loss is the mean NLL of the correct next characters.

7. **Training it**
   On all 228,146 bigrams: forward, set W.grad to None, loss.backward(), W += -50 * W.grad. A regulariser, 0.01 * (W**2).mean(), pulls W toward zero, where all logits are equal and predictions are uniform — the gradient-based twin of adding counts.

8. **Same destination**
   The trained net reaches a loss of about 2.48, the counts' level, and with the same random seed it samples almost exactly the same names. The difference is what comes next: the table cannot grow to longer contexts, and the net can.

WORKED EXAMPLE

Index the probabilities of the right answers in one step — probs[torch.arange(n), ys] — take the log, average and negate: that is the loss, for the count table and the network alike. Then compare it against the counts model's 2.48 as the target a one-character-context model can reach.

WATCH FOR

Dropping keepdims=True when normalising. P.sum(1) has shape 27 and broadcasts as a row, so each column is divided by a row's total. Nothing errors and the samples still look plausible; only the numbers are wrong.

CHECK YOURSELF · ANSWERS AT THE BACK

5.1 The first count table marks names with separate start and end tokens; the final one uses a single '.' at index 0 for both. Why is one token enough?

A Names never contain a dot, so the model cannot mistake it for a letter
B A start only ever opens a bigram and an end only closes one, so they never clash
C PyTorch needs the count table to be square, so the two markers had to be merged
D One shared token means every sampled name starts and ends with one letter

5.2 How many training bigrams does the name 'isabella' contribute, once it is wrapped in the start/end token?

Your answer: _____

5.3 P is 27 × 27. Why must each row be divided by P.sum(1, keepdims=True) rather than by P.sum(1)?

A keepdims makes the sum run over columns instead of rows, as is needed
B Without keepdims PyTorch raises an error, because the shapes cannot broadcast
C The two are the same; keepdims only changes how the result is printed
D Without keepdims the sums broadcast as a row, and columns get divided

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SPCH 100 · STORYTELLING · LECTURE 6 OF 22

What is a story — and why tell one

Mon 12 Oct · 18 min video · Watch on YouTube

THE TAKEAWAY

Dicks's case is that a told story is remembered, earns trust and changes the teller — and that telling one well is a matter of deciding what to say before you say it, not of having lived a dramatic life.

THE ARGUMENT · 6 STEPS

1. **A reluctant first night**
   Friends pushed him to his first Moth StorySLAM. He hoped his name would stay in the hat, his wife made him go up, and with the first words at the microphone he knew he wanted to come back.

2. **Stories are kept, data is not**
   Nobody asks to see a slide deck again, but people re-watch films they know by heart. Whatever you put inside a story gets remembered.

3. **Vulnerability buys connection**
   After a vulnerable story, strangers hand him their secrets — five women told him about miscarriages they had told no one else. Going first gives other people permission.

4. **Benevolent persuasion**
   He opened the talk with a story so the audience would like him before he asked them for anything. A story earns trust, and trust makes people follow.

5. **Personal understanding**
   He refused a Gatorade on a 100-degree golf course and only understood why when he asked himself: a hungry child learns never to accept food he cannot pay back.

6. **Most of us were trained to tell bad stories**
   Our audiences love us and put up with rambling. The fix is simple decision-making — think about what you will say before you say it — which anyone can learn.

WORKED EXAMPLE

When one of your own reactions seems out of proportion — refusing help, snapping at a stranger, avoiding a place — ask why you did it, then ask again. The answer that comes from years earlier is usually the story; the reaction is its doorway.

WATCH FOR

Believing you need a dramatic life to have stories. He has been through car crashes and jail and calls those the hardest to tell; what works is the small moment you would normally walk past.

CHECK YOURSELF · ANSWERS AT THE BACK

6.1 According to Dicks, what mainly separates a good storyteller from a bad one?

A Having lived a dramatic, eventful life
B Deciding what to say before saying it
C A trained performer's voice and timing
D Memorising the story word for word first

6.2 Why does he open the talk with his own story before giving any advice?

A To show the audience he has won enough contests to be worth hearing
B To warm the room up with a laugh before the serious advice starts
C So the audience likes and trusts him before he asks them to act
D To prove that only a hard life gives you stories worth telling

6.3 What did refusing a free drink on the golf course reveal when he asked himself why?

A A habit from a hungry childhood still ran his behaviour
B That he still felt he hadn't earned a place in that group
C That he quietly resented his friend's easy generosity
D That he was trying too hard to look independent and tough

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 7 OF 22

Limits, L'Hôpital's rule, and epsilon delta definitions

Tue 13 Oct · 18 min video · Watch on YouTube

THE TAKEAWAY

Limits exist to avoid talking about infinitely small changes, and once you have them the derivative can turn around and evaluate limits.

THE ARGUMENT · 6 STEPS

1 **The formal definition is what you have been doing all along**

$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$. The h is the same thing as dx — a concrete, finitely small number like 0.001. A different letter is used only to make clear that nothing infinitesimal is involved.

2 **Nothing here references an infinitely small change**

That is the point of limits. They let you ask what happens as a nudge shrinks without ever having to say what a nudge of size zero would mean. Reading dx as a real nudge is not a crutch; it is a translation of this definition.

3 **"Approaches" made airtight: epsilon and delta**

Pick any distance ε around the candidate value, however small. If a limit exists you can always find a range δ around the input so that every input within δ lands within ε. The order matters: ε first, then δ.

4 **And when the limit fails, it fails visibly**

A function jumping from 1 to 2 at the trouble point: shrink the input range as far as you like and the output range still straddles a gap of 1. No δ works for $\varepsilon = 0.4$, so no limit exists.

5 **$0/0$ is the interesting case, and derivatives crack it**

If $f(a) = g(a) = 0$, then near a each is approximately its derivative times dx. The ratio is $\dfrac{f'(a)\,dx}{g'(a)\,dx}$ and the dx cancel, leaving $f'(a)/g'(a)$.

6 **That is L'Hôpital's rule**

Differentiate top and bottom SEPARATELY — not as a quotient — and plug the trouble input back in. It cannot be used to discover derivative formulas, since it presupposes them; discovering those still takes the geometry.

WORKED EXAMPLE

$\lim_{x\to 1} \dfrac{\sin(\pi x)}{x^2 - 1}$. Both vanish at 1. The numerator changes by $\pi\cos(\pi)\,dx = -\pi\,dx$; the denominator by $2x\,dx = 2\,dx$. The dx cancel and the limit is exactly $-\pi/2$.

WATCH FOR

Applying L'Hôpital with the quotient rule. You differentiate numerator and denominator independently and then divide — the rule is about two separate derivatives, not the derivative of the fraction.

CHECK YOURSELF · ANSWERS AT THE BACK

7.1 A limit fails to exist, in epsilon-delta terms, when:

A The function is undefined at the point
B Some ε exists for which no δ is small enough to trap the outputs
C The graph is steep
D The derivative is zero

7.2 By L'Hôpital's rule, $\lim_{x\to 2} \dfrac{x^2 - 4}{x - 2}$ equals:

Your answer: _____

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 8 OF 22

Integration and the fundamental theorem of calculus

Tue 13 Oct · 21 min video · Watch on YouTube

THE TAKEAWAY

The integral is the limit of a sum of thin rectangles, and you evaluate it by finding an antiderivative and looking at just two inputs.

THE ARGUMENT · 6 STEPS

1. **Constant velocity would be easy, so pretend it is**
   Distance is velocity times time, and on a velocity-against-time plot that product is literally an AREA. Chop $[0, 8]$ into intervals of width dt and treat the speed as constant across each one.

2. **Which choice of constant does not matter**
   Anywhere between the interval's start and end speed will do. What matters is only that the approximation improves as dt shrinks — that pretending the car jumps between constant speeds becomes less wrong the finer the chopping.

3. **$\int_0^8 v(t)\,dt$ says exactly that**
   The stretched S is a sum, $v(t)\,dt$ is one rectangle, and dt does double duty: it is a factor in each term AND the spacing between samples. Shrink it and each rectangle shrinks while their number grows.

4. **Let the upper bound vary and you get a function**
   $s(T) = \int_0^T v(t)\,dt$ is the distance travelled after T seconds. Nudge T by dT and the area grows by a sliver of height $v(T)$ and width dT — so $ds/dT = v(T)$.

5. **So integrating is undoing a derivative**
   For $v = 8t - t^2$, ask what has that as its derivative: $4t^2$ gives $8t$, and $-\frac{1}{3}t^3$ gives $-t^2$. Any constant may be added, since constants differentiate to zero — hence the $+C$.

6. **The fundamental theorem, and why it feels like cheating**
   $\int_a^b f = F(b) - F(a)$. Subtracting at the lower bound is what forces the integral from a to a to be zero, and it cancels whatever C you chose. A sum over a whole continuum, computed from two numbers.

WORKED EXAMPLE

Integrals measure SIGNED area. Where the graph dips below the axis, $v(t)$ is negative, so $ds = v\,dt$ is negative and the car moves backwards. That area subtracts — which is why the result is displacement rather than distance travelled.

WATCH FOR

Forgetting to subtract the antiderivative at the lower bound. $F(b)$ alone is only correct when $F(a)$ happens to be zero; otherwise the integral from a to a would not come out zero.

CHECK YOURSELF · ANSWERS AT THE BACK

8.1 The derivative of the area-so-far function equals the height of the graph because:

A Areas and heights are the same units
B Nudging the right edge by dT adds a sliver that is nearly a rectangle of height $f(T)$ and width dT
C The graph is a straight line
D Integrals and derivatives are unrelated

8.2 With $v(t) = t(8 - t)$, the distance travelled between $t = 0$ and $t = 6$ is:

Your answer: _____

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 9 OF 22

What does area have to do with slope?

Tue 13 Oct · 13 min video · Watch on YouTube

THE TAKEAWAY

Averaging a continuous quantity is an integral divided by a width — and reading that as an average slope explains why antiderivatives solve integrals at all.

THE ARGUMENT · 6 STEPS

1. **You cannot add infinitely many values and divide by infinity**
   But you can sample finitely many, average those, and ask what happens as the sample grows. Whenever you want to add up values across a continuum and it does not make sense, the answer is almost always an integral.

2. **Rewrite the finite average in terms of the spacing**
   With samples spaced dx apart across $[0, \pi]$, there are about π / dx of them. Dividing the sum of heights by that count is the same as multiplying the sum by dx and dividing by π.

3. **Which turns the numerator into an integral**
   The terms become $\sin(x)\,dx$ — areas, not heights. So the average height is $\frac{1}{\pi}\int_0^{\pi} \sin x\,dx$: area divided by width. Even the units agree.

4. **Evaluate it the usual way**
   The antiderivative of $\sin$ is $-\cos$, whose slope does visibly track the sine graph's height. Between 0 and π its value rises by exactly 2, so the area is 2 and the average height is $2/\pi \approx 0.64$.

5. **Now read $\frac{F(b) - F(a)}{b - a}$ as a slope**
   Rise over run on the antiderivative's graph, between the two endpoints. That is the second perspective the chapter is for.

6 **And f IS the slope of F at every point**

So the average value of f is the average of all those tangent slopes — and the average slope across a stretch of graph is just the overall slope from start to end. That is why comparing two endpoints suffices.

WORKED EXAMPLE

A second cue for reaching for an integral. The first was: the problem breaks into a sum of many small things. The second is: you understand an idea in a finite setting — an average, a count, an expectation — and want it over a continuous range. Probability is full of this.

WATCH FOR

Dividing the integral by the number of samples. The sample count has already been absorbed: once the terms carry dx, what you divide by is the WIDTH of the interval, not how many points you took.

CHECK YOURSELF · ANSWERS AT THE BACK

9.1 The average value of f on $[a, b]$ equals $\frac{F(b)-F(a)}{b-a}$, which is also:

A The maximum of f
B The slope of the antiderivative's graph between the two endpoints
C The area under F
D The derivative of f

9.2 The area under $\sin x$ from 0 to π is:

Your answer: _____

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 10 OF 22

Higher order derivatives

Tue 13 Oct · 6 min video · Watch on YouTube

THE TAKEAWAY

The second derivative is the derivative of the derivative — visually it is curvature, and physically it is acceleration.

THE ARGUMENT · 4 STEPS

1 **It measures how the slope is changing**

Curving upwards means the slope is increasing, so the second derivative is positive. Curving downwards means it is negative. Where the graph is locally straight, there is no curvature and the second derivative is 0.

2 **Sign and size are different questions**

Two graphs can both curve upwards at $x = 4$ and have very different second derivatives — one where the slope rockets up, one where it creeps. The sign says which way it bends; the magnitude says how sharply.

3 **Where the notation comes from**

Take two steps of size dx. The first changes f by df_1, the second by df_2, and the difference between those changes is ddf — proportional to dx^2, so about 0.0001 when $dx = 0.01$. The second derivative is $ddf/(dx)^2$, written $\frac{d^2f}{dx^2}$.

4 **The most visceral reading is acceleration**

Distance, then velocity, then acceleration. A positive second derivative is the feeling of the seat pushing you forward; a negative one is slowing down. And the third derivative, genuinely, is called jerk.

WORKED EXAMPLE

This is the setup for Taylor series. Approximating a function near a point uses the value, then the slope, then the curvature, then the next order up — each higher derivative pinning down one more aspect of the shape.

WATCH FOR

Reading the 2s in $\frac{d^2f}{dx^2}$ as ordinary squares. The top counts how many times you differentiated; the bottom really is $(dx)^2$, because the change-in-the-change is second order in the step.

CHECK YOURSELF · ANSWERS AT THE BACK

10.1 A graph curving downwards at a point has:

A A positive second derivative
B A negative second derivative, because the slope is decreasing there
C A zero first derivative
D No second derivative

10.2 For $f(x) = x^3$, the second derivative at $x = 3$ is:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 11 OF 22

The five-second moment: the instant a story is about

Tue 13 Oct – Wed 14 Oct · day 1–2 of 2 · 24 min video · Watch on YouTube

THE TAKEAWAY

A story is about the instant someone changes — usually small and internal — and the dramatic events around it only matter as the road to that instant.

THE ARGUMENT · 6 STEPS

1. **The crash that is not the story**
   At seventeen, two days before Christmas, he goes through a windshield after a day spent buying surprise presents for friends; his heart stops in the ambulance.

2. **The doorway that is**
   His parents go to check the car before they come to him. His friends, alerted by a McDonald's phone tree, crowd the emergency-room doors chanting his name. They gave him the gift of family.

3. **Why nobody cries at the death**
   Audiences blink when he dies and cry when his friends arrive. Nobody can connect to a windshield; everybody can connect to finding out they are loved.

4. **Realization or transformation**
   A story is change over time: either I now see something differently, or I have become someone different. The change itself takes an instant.

5. **Small is enough**
   Most of his stories are moments an outsider would not notice, because most change happens inside your head. A small change of mind or heart is enough.

6. **The bench and the fountain**
   His son promised to build him a fountain when he dies. Three months later he realised that playing golf with his son was the one childhood promise to himself he had kept.

WORKED EXAMPLE

Write the moment as 'before, I thought X; in that instant, I thought Y'. Then work backwards: what does the audience need to know for Y to land? Anything that does not serve that instant can go — including the drama.

WATCH FOR

Mistaking the biggest event for the story. The crash is spectacular and inert; the instant of feeling loved is small, and it is what the audience actually responds to.

CHECK YOURSELF · ANSWERS AT THE BACK

11.1 In the car-crash story, what is the five-second moment?

A Going through the windshield two days before Christmas
B His heart stopping in the ambulance, on the way to the hospital
C Realising, as his friends crowd the doorway, that he is loved
D His parents checking the car before they came in to see him

11.2 Why do audiences not cry when he describes dying?

A Nobody can connect to the experience of going through a windshield
B He hurries through that part so it doesn't upstage the ending
C The details are too graphic, so listeners pull back from them
D They know he survived, so his death holds no real suspense for them

11.3 What did the golf-bench moment finally mean to him?

A That his son wanted more time together than golf could give
B That he should give up golf and spend weekends with his son
C That he still resented the way his own father had raised him as a boy
D That playing golf with his son kept a promise he'd made as a boy

MATH 130 · PROBABILITY · LECTURE 12 OF 22

Birthday Problem, Properties of Probability

Wed 14 Oct · 49 min video · Watch on YouTube

THE TAKEAWAY

Twenty-three people suffice for a better-than-even birthday match, because the quantity that matters is not k but the $\binom{k}{2}$ PAIRS — and inclusion-exclusion is the general tool for the probability of a union.

THE ARGUMENT · 13 STEPS

1. **State the assumptions before computing anything**
   365 days, equally likely, births independent. None is exactly true — there are real seasonal effects, and they differ by country — but each has to be said out loud, because the naive definition depends on them.

2. **Above 365 people the answer is 1**
   More people than days means some day holds two, by the pigeonhole principle. Worth stating: it is the sanity check at the extreme end.

3 **Compute the complement**
$P(\text{no match}) = \dfrac{365 \cdot 364 \cdots (365-k+1)}{365^k}$, then subtract from 1. Count the terms carefully — the last is $365-k+1$, and being off by one here is the classic error.

4 **The numbers**
$k = 23$ gives 50.7%; $k = 50$ gives 97%; $k = 100$ gives over 99.999%. Almost everyone guesses something above 100 for the first one.

5 **Why 23 is not absurd**
$\binom{23}{2} = 253$ pairs. You are not asking about one coincidence but about 253 chances at a coincidence, and 253 is the same order as 365. Looking at k is intuitive and is the wrong quantity.

6 **The same question, one day apart, needs only 14**
And proving THAT is much harder — you cannot just replace 364 by 362, because the excluded days overlap as the year fills up. Approximation methods later in the course handle it.

7 **The complement rule, proved from the axioms**
$S = A \cup A^c$, disjoint, so $1 = P(A) + P(A^c)$. Two lines from axioms 1 and 2.

8 **Monotonicity**
If $A \subseteq B$ then $P(A) \le P(B)$: write $B = A \cup (B \cap A^c)$, disjoint, so $P(B) = P(A) + \text{something} \ge 0$.

9 **Disjointification**
Axiom 2 only applies to disjoint events, so make them disjoint: $A \cup B = A \cup (B \cap A^c)$. That single move is behind both of the proofs above and the next one.

10 **The union of two events**
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Add both areas, then remove the overlap you counted twice.

11 **Inclusion-exclusion in general**
Add the singles, subtract the pairs, add the triples, alternating. For three events the triple intersection is added three times and subtracted three times, so it must be added back once — which is exactly what the pattern does.

12 **De Montmort's matching problem**
Shuffle cards numbered 1 to n and count as you turn them over; you win if some card's position equals its number. $P(A_j) = 1/n$ by symmetry, $P(A_1 \cap \cdots \cap A_k) = (n-k)!/n!$, and inclusion-exclusion collapses almost everything.

13 **And the answer is $1 - 1/e$**
What survives is $1 - \frac{1}{2!} + \frac{1}{3!} - \cdots$, the Taylor series for e^{-1}. About 0.632, and — remarkably — it barely changes with n. This constant will keep appearing.

WORKED EXAMPLE

Ask whether the event or its complement is easier to count. 'No two share a birthday' is a clean product; 'at least one pair shares' is a mess of overlapping cases. The complement rule makes that a free choice.

WATCH FOR

Trusting intuition about coincidences. The biggest coincidence would be no coincidences at all: count how many opportunities there are for the event, not how many objects there are.

CHECK YOURSELF · ANSWERS AT THE BACK

12.1 Why does 23 people suffice for a better-than-even birthday match?

A Because $365/23 \approx 16$
B Because there are $\binom{23}{2} = 253$ pairs, and 253 is comparable to 365
C Because birthdays are not really uniform
D Because of the pigeonhole principle

12.2 The probability of a birthday match among 23 people, to three decimal places, is:

Your answer: _____

12.3 The probability of at least one match in de Montmort's card game with 52 cards, to three decimal places, is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 13 OF 22

Makemore 2 — MLP, following Bengio et al. 2003

Wed 14 Oct – Thu 15 Oct · day 1–2 of 2 · 76 min video · Watch on YouTube

THE TAKEAWAY

To see more than one character back, stop counting and learn: embed each character as a small vector, concatenate the last three, and feed them through a hidden layer — Bengio et al.'s MLP — trained on minibatches and judged on data it never trained on.

THE ARGUMENT · 7 STEPS

1. **A context of three characters**
   block_size = 3: each example is the previous three characters (padded with '.' at the start of a name) and the next character as the label. A name of n letters gives n + 1 examples.

2. **Train, dev and test**
   Shuffle the words and split 80 / 10 / 10 — 182,441, 22,902 and 22,803 examples. Train on the first, tune choices on the second, and keep the third for the end.

3 **An embedding table**

C maps each of the 27 characters to a learned vector — 2 numbers at first, so the embedding can be plotted, then 10. C[X] looks up every character of every context at once.

4 **Concatenate, then a hidden layer**

emb.view(-1, 30) lays each context's three 10-dimensional embeddings side by side as one 30-long vector; a 200-unit tanh layer and a 27-way output layer follow. 11,897 parameters in all, and F.cross_entropy computes the loss from the logits.

5 **Minibatches**

Each step uses 32 random training examples, not all 182,441: the gradient is noisier but far cheaper, so the model can take hundreds of thousands of steps.

6 **Finding a learning rate**

Sweep candidate rates evenly in the exponent — 10 to the power of linspace(-3, 0) — and watch the loss to find a good one. The final run uses 0.1 for 100,000 steps, then decays to 0.01.

7 **Judge it on the dev set**

Training loss 2.126, dev loss 2.170: close, so the model is not simply memorising, and it beats the bigram model's 2.48. Samples ('kaleigh.', 'quinton.', 'jace.') now look much more like names.

WORKED EXAMPLE

Write every tensor's shape next to the line that makes it: X is (N, 3), C[X] is (N, 3, 10), the view is (N, 30), the hidden layer (N, 200), the logits (N, 27). Most bugs in this model are a shape that is not what you thought, and the comments in the notebook do exactly this.

WATCH FOR

Reporting the training loss as the model's quality. Only the dev split says how it does on names it has not seen; and the test split is spent once, at the end, or it stops being an honest measure.

CHECK YOURSELF · ANSWERS AT THE BACK

13.1 C is 27 × 10, W1 is 30 × 200 with a bias of 200, W2 is 200 × 27 with a bias of 27. How many parameters does the MLP have? (digits only, no commas)

Your answer: _____

13.2 With block_size = 3, how many (context, next character) examples does the name 'emma' produce?

Your answer: _____

13.3 Why does the forward pass reshape the embeddings with emb.view(-1, 30)?

A To average the three characters' embeddings into a single vector
B To lay the three 10-number embeddings side by side as one input
C To drop the batch dimension so the matrix multiply can broadcast
D To sort the context characters by index before the hidden layer

MATH 110 · LINEAR ALGEBRA · LECTURE 14 OF 22

Abstract vector spaces

Thu 15 Oct · 17 min video · Watch on YouTube

THE TAKEAWAY

A vector is anything you can sensibly add and scale, and every result in linear algebra applies to all of them at once.

THE ARGUMENT · 6 STEPS

1. **The question the series opened with**
   Is a vector an arrow, or a list of numbers? Once you are fluent with changing basis, coordinates start to feel arbitrary — and determinants and eigenvectors plainly do not care which basis you picked.

2. **Functions add and scale**
   $(f+g)(x) = f(x) + g(x)$, and scaling a function multiplies every output. That is adding and scaling coordinate by coordinate, exactly as with arrows — there are just infinitely many coordinates.

3. **So functions have linear transformations too**
   The derivative is one. It is additive and it respects scaling, which is the pair of facts every calculus student uses daily without ever hearing them called linearity.

4. **The derivative, written as a matrix**
   Take polynomials with basis $1, x, x^2, x^3, \ldots$. A polynomial becomes a coordinate list ending in zeros, and the derivative becomes an infinite matrix carrying $1, 2, 3, \ldots$ down an offset diagonal.

5. **Built the usual way**
   Each column is the derivative of one basis function, written in coordinates. That is the same rule as for arrows: a linear map is determined by where the basis goes.

6. **The axioms are an interface, not a foundation**
   Eight rules that adding and scaling must obey. Satisfy them and every theorem applies to your objects, whatever they are. That is why textbooks define linearity as additivity and scaling rather than by grid lines.

WORKED EXAMPLE

Differentiate $x^3 + 5x^2 + 4x + 5$ by matrix. Its coordinates are $(5, 4, 5, 1, 0, \ldots)$. The matrix returns $(4, 10, 3, 0, \ldots)$, which reads back as $3x^2 + 10x + 4$ — the derivative, obtained by matrix-vector multiplication.

WATCH FOR

Treating the abstract definition as the real one and the arrows as a crutch. The arrows are where the intuitions come from; the axioms are how they travel.

KEY CONCEPTS

A basis is the ruler you measure with

CHECK YOURSELF · ANSWERS AT THE BACK

14.1 The eight vector-space axioms exist to:

A Define what an arrow is
B Let a result proved once apply to every set that satisfies them
C Make determinants computable
D Guarantee that a basis is finite

14.2 In the basis $1, x, x^2, x^3$, the derivative of $x^3 + 5x^2 + 4x + 5$ has coefficient of x equal to:

Your answer: _____

MATH 110 · LINEAR ALGEBRA · LECTURE 15 OF 22

The Geometry of Linear Equations

Thu 15 Oct · 40 min video · Watch on YouTube

THE TAKEAWAY

Every linear system has a row picture and a column picture, and the column picture is the one that survives into higher dimensions.

THE ARGUMENT · 5 STEPS

1. **One problem, written three ways**
   n equations in n unknowns. The row picture takes one equation at a time; the column picture takes one column at a time; the matrix form $A\vec{x} = \vec{b}$ packages both. Strang stars the column picture, and the lecture earns the star.

2. **The row picture is where they meet**
   Each equation draws a line in two dimensions, a plane in three. The solution is the point they all share — two planes meet along a line, and the third plane cuts that line at a point.

3. **The column picture asks for a combination**
   $A\vec{x} = \vec{b}$ is asking: how much of each column do I need so that they add up to $\vec{b}$? That linear combination of columns is the single most fundamental operation in the course.

4. **Why the column picture scales and the row picture does not**
   Three planes intersecting is already hard to see. Nine is hopeless. But combinations of nine vectors in nine dimensions is still a thought you can hold, even without picturing it.

5. **Solvable for every $\vec{b}$ is a question about span**
   Ask whether the combinations of the columns fill the whole space. If the third column lies in the plane of the first two, it adds nothing, and most right-hand sides are simply unreachable.

WORKED EXAMPLE

$2x - y = 0$ and $-x + 2y = 3$. In the column picture: find x, y with $x(2, -1) + y(-1, 2) = (0, 3)$. One of the first column plus two of the second gives $(2 - 2,\ -1 + 4) = (0, 3)$. So $x = 1$, $y = 2$.

WATCH FOR

Reading $A\vec{x}$ only as rows dotted with $\vec{x}$. It is equally a combination of the columns, and that is the reading the rest of the course is built on.

KEY CONCEPTS

Span is everywhere you can reach

CHECK YOURSELF · ANSWERS AT THE BACK

15.1 $A\vec{x} = \vec{b}$ has a solution for EVERY $\vec{b}$ exactly when:

A A is square
B Combinations of the columns fill the whole space
C The rows are perpendicular
D $\vec{b}$ is the zero vector

15.2 Combining $(2, -1)$ and $(-1, 2)$ with $x = 1$ and $y = 2$, the second component is:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 16 OF 22

Anecdote, bait and reflection — the two building blocks

Thu 15 Oct · 5 min video · Watch on YouTube

THE TAKEAWAY

Ira Glass reduces a story to two building blocks — the anecdote, which carries momentum, and the moment of reflection, which carries meaning — and says the craft is weaving them together while baiting the listener with questions.

THE ARGUMENT · 5 STEPS

1 **Not the school essay**
We were taught a topic sentence followed by supporting facts. Broadcast stories work on two different blocks: the anecdote and the reflection.

2 **The anecdote is a sequence of actions**
This happened, which led to this, which led to this. Even the dullest sequence — a man wakes, the house is too quiet, he walks downstairs — feels like a train with a destination.

3 **Bait**
The quiet house raises a question, and the listener assumes it will be answered. Keep raising and answering questions from the first line.

4 **The moment of reflection**
At some point someone has to say why you are listening — the bigger thing the story is driving at.

5 **Two ways to fail**
A thrilling anecdote that means nothing, or a smart idea with no story under it. You need both, and a good piece flips back and forth between them.

WORKED EXAMPLE

Draft in alternation: a little action, a line of meaning, more action, more meaning. If the meaning lines could be deleted without anyone noticing, they are not reflections yet; if the action lines are summaries, they are not anecdote.

WATCH FOR

Falling in love with a great anecdote that predictably leads nowhere. Glass calls that the bane of his existence — and says you must be ruthless enough to see that one of the two blocks is missing.

CHECK YOURSELF · ANSWERS AT THE BACK

16.1 What are Ira Glass's two building blocks of a story?

A The setup and the punchline that pays it off
B The anecdote and the moment of reflection
C The character and the problem they must solve
D The hook and the question it leaves open

16.2 In his example of a man waking in a quiet house, what is the 'bait'?

A The unusual quiet, which raises the question why
B The man waking up, which tells us who the story is about
C The staircase, which promises movement and a destination
D The early hour, which suggests something has gone wrong

16.3 Which failure does he call the bane of his existence?

A A beautiful reflection with no anecdote to carry it
B A story that answers its questions too early
C A thrilling anecdote that means absolutely nothing
D Interviewees who explain the point instead of showing it

SPCH 100 · STORYTELLING · LECTURE 17 OF 22

The elephant: a reason to listen in the first thirty seconds

Thu 15 Oct · 10 min video · Watch on YouTube

THE TAKEAWAY

Dicks's essential stake is the elephant: an early, clear signal of what the story is about and what could go wrong, which gives the audience a reason to stay — and which may turn into a different stake as the story develops.

THE ARGUMENT · 6 STEPS

1. **What stakes do**
   They make an audience worry, wonder, root for you or hope for your demise. They are what drives people to the end of a story.

2. **Why a speaker needs it more than a film**
   Films have trailers, so we walk in knowing the stakes. When you start to speak, nobody knows what you are about to say.

3. **The spoon of power**
   A boy pulls a spoon out of a leaf pile, Dicks declares it the spoon of power, and now he must have it. Will he get the spoon? That question carries the story.

4. **No petrol, no money**
   His tyre disintegrates a hundred miles from home; seven hours later all his money is in a stranger's hand and the tank is empty. How will he get home?

5. **The elephant changes colour**
   On a widower's porch, lying about a charity, getting home stops mattering. The new stake is how he will face what he has done.

6. **Beyond stories**
   A pitch, a keynote, a product launch all need one. 'I have been waiting two and a half years to tell you this' is an elephant.

WORKED EXAMPLE

In the first thirty seconds, state the situation that creates the question, not a summary of the story: where you are, what you want, what is in the way. Every later sentence then reads as a step toward the answer.

WATCH FOR

A cheap hook — a shocking first line unrelated to what the story is really about. The elephant has to be the question the story actually answers, at least for now.

CHECK YOURSELF · ANSWERS AT THE BACK

17.1 Why does a spoken story need an elephant more than a film does?

A Films can show the danger, while a speaker can only describe it
B Films have trailers, so their audience already knows the stakes
C Live audiences are less patient than people sitting in a cinema
D A spoken story is shorter, so it has less time to build stakes

17.2 What does it mean that an elephant can change colour?

A The narrator can switch from one character's view to another's
B The tone can darken as the story goes on, from comic to sad
C The ending can reveal that the stake was never real in the first place
D The main stake can shift as the story turns, as on the widower's porch

17.3 Which of these openings is an elephant?

A My tyre has just disintegrated, I have no phone and no money, and I am a hundred miles from home.
B This is a story about loneliness, and about a stranger who taught me what it means.
C I want to tell you about something that happened to me once, years ago, on a very long drive home.
D Let me start with some background about my family, my car, and why I was on that road at all.

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 18 OF 22

Taylor series

Fri 16 Oct · 22 min video · Watch on YouTube

THE TAKEAWAY

Taylor series translate derivative information at a single point into approximation information near that point, by matching one derivative per coefficient.

THE ARGUMENT · 6 STEPS

1 **Build the approximation one derivative at a time**
To fit $c_0 + c_1x + c_2x^2$ to $\cos x$ near 0: c_0 makes the values agree, c_1 makes the slopes agree, c_2 makes the curvatures agree. Each constant is responsible for exactly one derivative, and nothing else.

2 **Which is why plugging in $x = 0$ matters so much**
Every higher-order term carries an x, so at $x = 0$ it washes out of every derivative below its own order. Adding a new term never disturbs the coefficients already chosen — that independence is what makes the construction work.

3 **Factorials appear because the power rule cascades**
Differentiating x^n n times leaves $1 \cdot 2 \cdots n$. So you do not set the coefficient equal to the derivative you want; you divide by $n!$ to cancel that. Hence $c_n = f^{(n)}(0)/n!$.

4 **Cosine's derivatives cycle, so its coefficients do**
$\cos, -\sin, -\cos, \sin$, repeating; at 0 that is $1, 0, -1, 0$. So the polynomial is $1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots$, and $1 - \frac{x^2}{2}$ is the best CUBIC approximation too, since c_3 comes out zero.

5 **The second-order term is a triangle**
Approximate an area function near a: the rectangle $f(a)(x-a)$ plus the triangle on top of it, base $x - a$ and height $f'(a)(x-a)$, giving $\frac{1}{2}f'(a)(x-a)^2$. Every term in the formula is a piece of that picture.

6 **Infinitely many terms is a series, and it may not converge**
e^x equals its Taylor series everywhere. $\ln x$ expanded at 1 converges only for inputs in $(0, 2)$ — outside that the partial sums thrash about. The width of the good region is the radius of convergence.

WORKED EXAMPLE

To approximate near some a other than 0, write the polynomial in powers of $(x-a)$ and evaluate every derivative at a. It looks heavier, but all it does is make a behave the way 0 did — so plugging in $x = a$ still kills everything but one term.

WATCH FOR

Assuming a Taylor series is valid wherever the function is. $\ln x$ is perfectly well defined at $x = 3$, but the derivative information gathered at $x = 1$ does not reach that far, and the series diverges.

CHECK YOURSELF · ANSWERS AT THE BACK

18.1 Taylor coefficients are divided by $n!$ because:

A Factorials make the numbers smaller
B Differentiating x^n n times produces a factor of $n!$, which must be cancelled
C The series would otherwise be infinite
D It makes the graph symmetric

18.2 Using $\cos x \approx 1 - \frac{x^2}{2}$, the estimate of $\cos(0.1)$ is:

Your answer: _____

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 19 OF 22

The other way to visualize derivatives

Fri 16 Oct · 14 min video · Watch on YouTube

THE TAKEAWAY

Read a function as a transformation of the number line and the derivative becomes a local stretch factor — a view that survives into contexts where graphs do not.

THE ARGUMENT · 6 STEPS

1. **The slope reading is one view, not the definition**
   What a derivative is really about is how sensitive the output is to a tiny nudge in the input. Slope is what that sensitivity looks like when you draw a graph — and graphs stop being drawable once inputs and outputs are not plain numbers.

2. **So map the input line onto an output line instead**
   Zoom in near an input and look at evenly spaced points. The derivative is the factor by which that cluster gets stretched or squeezed on arrival. For x^2 near 1 the spacing roughly doubles; near 3 it multiplies by about 6.

3. **Every value of the derivative has a picture**
   Less than 1 in magnitude means contraction. Zero means the neighbourhood collapses towards a point as you zoom further. Negative means the cluster is also FLIPPED — near -2, x^2 acts locally like multiplying by -4.

4. **Which explains a puzzle about $1 + \frac{1}{x}$**
   Its fixed points are $\varphi \approx 1.618$ and $-1/\varphi \approx -0.618$. Both satisfy the equation, yet hitting enter repeatedly on a calculator always lands on φ, from any starting number.

5. **Because stability is a statement about the derivative**
   Near φ the derivative has magnitude about 0.38, so each pass pulls the neighbourhood inward — a gravitational tug. Near the little brother it exceeds 2, so points are flung away. Stable and unstable fixed points.

6. **The rule worth keeping**
   A fixed point is stable when $|f'|$ there is less than 1 and unstable when it is greater. The sign only tells you whether the approach spirals or marches; the magnitude decides the fate.

WORKED EXAMPLE

This is the real reason to carry the view forward. Picturing a whole function as a transformation is clunkier than a graph for single-variable work, but multivariable calculus, complex analysis and differential geometry all live in places where no graph can be drawn.

WATCH FOR

Judging a fixed point by the derivative's sign. A derivative of -0.38 is stable and -2 is not, even though both are negative — it is the magnitude against 1 that decides.

CHECK YOURSELF · ANSWERS AT THE BACK

19.1 A fixed point of f is stable under repeated application when:

A f' is positive there
B $|f'|$ there is less than 1, so nearby points contract towards it
C f' is zero there
D f is increasing

19.2 Read as a transformation, x^2 stretches a small cluster of points near the input 3 by a factor of:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 20 OF 22

Makemore 3 — activations, gradients, BatchNorm

Fri 16 Oct – Sun 18 Oct · day 1–2 of 3 · 116 min video · Watch on YouTube · finishes in week 3

THE TAKEAWAY

A deep net trains well only if its activations and gradients stay well scaled from the first step: start with an output that is unconfident, hidden units that are not saturated, weights scaled by fan-in — and batch normalisation to hold every layer's pre-activations near unit Gaussian — then watch the statistics instead of guessing.

THE ARGUMENT · 8 STEPS

1. **The loss log**
   The notebook keeps a running log as each fix lands: original (train 2.125, val 2.168); softmax confidently wrong fixed (2.07 / 2.13); tanh too saturated fixed (2.036 / 2.103); a principled Kaiming init (2.038 / 2.107); batch norm added (2.067 / 2.105).

2 **Don't start confidently wrong**

W2 is scaled by 0.01 and b2 set to 0, so the first logits are near zero and the first predictions near uniform. The first loss is 3.32 — about $\ln 27$, what a model that knows nothing should score — instead of a large loss spent in the first steps just un-learning random confidence.

3 **Don't start saturated**

If the hidden pre-activations are large, tanh outputs pile up at ±1, where its local gradient $1 - t^2$ is about zero and the unit passes almost nothing backward. Scaling W1 down keeps them in tanh's active range.

4 **Kaiming init instead of a magic number**

Scale W1 by gain / $\sqrt{\text{fan_in}}$, with gain 5/3 for tanh: randn * (5/3) / sqrt(30). The Linear class later uses / fan_in**0.5 by default.

5 **Batch normalisation**

Standardise each hidden unit's pre-activation over the batch — subtract the batch mean, divide by the batch std — then multiply by a learned gain and add a learned bias, so the layer can still choose its own scale and offset. A bias before batch norm is cancelled by the mean subtraction, so b1 and the Linear layers' biases are dropped.

6 **Batch norm at inference**

One example has no meaningful batch statistics, so batch norm keeps running estimates — bnmean_running and bnstd_running, nudged with momentum 0.001 every step (or calibrated over the whole training set afterwards) — and uses them when evaluating or sampling.

7 **PyTorch-style modules, deeper**

Linear, BatchNorm1d (with training mode, eps, momentum and running_mean / running_var buffers) and Tanh, stacked six layers deep — 47,024 parameters — with the last layer's gamma scaled by 0.1 to make the output less confident.

8 **Diagnostics to read**

Per tanh layer: mean, std and the share saturated ($|t| > 0.97$) of activations — about 0.64 std and 2% saturated here; the same for their gradients; each weight's gradient std relative to its data; and, over training, log10 of (lr × gradient std) / (data std), which should sit near -3.

WORKED EXAMPLE

Before a long run, check the first step. The initial loss should be about $\ln(\text{vocab})$; the hidden activations should have a std near the same value in every layer with few saturated; and the update-to-data ratios should sit around 10^{-3}. Each of these is one print statement, and each catches a problem that otherwise costs a full training run.

WATCH FOR

Evaluating a batch-norm network one example at a time with its training-mode statistics. A single example's mean is itself and its std is undefined, so the output is meaningless — switch the layers out of training mode so they use their running mean and variance.

CHECK YOURSELF · ANSWERS AT THE BACK

20.1 Why does the notebook scale W2 by 0.01 and set b2 to 0 at initialisation?

A To make the output layer train faster than the hidden layers
B To stop the embeddings in C from drifting during the first steps
C To keep the weights small enough that no regularisation is needed
D So the first predictions are near uniform, not confidently wrong

20.2 W1 is initialised as randn * (5/3) / sqrt(fan_in), with fan_in = n_embd × block_size = 10 × 3. What standard deviation is that? Three decimals.

Your answer: _____

20.3 The diagnostics print the share of tanh outputs with |t| above 0.97 as 'saturated'. Why does the lecture want that share small?

A Saturated units output exactly zero, so the forward pass loses them
B There tanh's local gradient, $1 - t^2$, is near zero, so little flows back
C Saturated units make the loss negative, which breaks cross-entropy
D Saturation only matters for the output layer, never for the hidden layers

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MATH 130 · PROBABILITY · LECTURE 21 OF 22

Conditional Probability

Sun 18 Oct · 50 min video · Watch on YouTube

THE TAKEAWAY

Conditional probability is $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$ — restrict the world to B and renormalise — and conditioning is the soul of statistics, because it is how beliefs get updated by evidence.

THE ARGUMENT · 13 STEPS

1. **The matching problem, finished**
   There are $\binom{n}{k}$ intersections of size k, each of probability $\frac{(n-k)!}{n!}$, and the product collapses to $\frac{1}{k!}$. So $P(\text{match}) = 1 - \frac{1}{2!} + \frac{1}{3!} - \cdots$ and $P(\text{no match}) \to e^{-1} \approx 0.37$.

2. **And it converges absurdly fast**
   Even at $n = 10$ the answer agrees with $1/e$ to about 10^{-8}. Two competing forces — each card is unlikely to match, but there are many cards — balance out at a constant, which is not what most people guess.

3. **Independence means multiply**
   A and B are independent when $P(A \cap B) = P(A)P(B)$. Nothing about vagueness like 'one tells you nothing about the other' — that is the intuition, this is the test.

4 **Independent is not disjoint, and confusing them is a disaster**
Disjoint means that if A occurs then B CANNOT. That is the opposite of independence, which says A occurring tells you nothing about B.

5 **Three events need four equations, not three**
All three pairs independent — pairwise independence — does NOT imply $P(A \cap B \cap C) = P(A)P(B)P(C)$, and neither implies the other. For n events, every subset must factor.

6 **Newton and Pepys**
At least one six in 6 dice, at least two in 12, or at least three in 18 — which is likeliest? Pepys believed the third and bet on it. The answer is the first: 0.665, 0.619, 0.597.

7 **'At least one' means take the complement**
$P(\text{at least one six in } 6) = 1 - (5/6)^6$. A union invites inclusion-exclusion, but its complement is an intersection of independent events, which is just a product.

8 **The binomial probability, derived on the spot**
$P(\text{exactly } k \text{ sixes in } n) = \binom{n}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{n-k}$: choose which positions hold the sixes, then fix each die. Worth deriving rather than memorising, because it will recur constantly.

9 **Newton's arithmetic was right and his intuition was wrong**
Stigler showed the intuitive argument had to be wrong WITHOUT decoding it: nothing in it used the fairness of the dice, yet with unfair dice the ordering can reverse. An argument that is insensitive to something the answer depends on cannot be correct.

10 **Pebble world**
Draw the sample space as pebbles whose masses sum to 1. Conditioning on B means deleting every pebble outside B and then RENORMALISING so the remaining masses sum to 1 again — which is exactly the division by $P(B)$.

11 **Frequentist world**
Repeat the experiment many times, circle the repetitions where B happened, and ask what fraction of THOSE also had A. Same formula, a different picture of what it means.

12 **Three theorems, all one line**
$P(A \cap B) = P(B)P(A \mid B) = P(A)P(B \mid A)$; the chain rule $P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2 \mid A_1) \cdots$; and Bayes' rule $P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$. The proofs are trivial; the implications have kept statisticians arguing for two centuries.

13 **The chain rule can be applied in any order**
There are $n!$ orderings, and they are all valid. On a given problem one may be routine and another intractable, so the ordering is a choice worth making deliberately.

WORKED EXAMPLE

When an event says 'at least one', reach for the complement before reaching for inclusion-exclusion. The complement of a union is an intersection, and if the pieces are independent an intersection is a product.

WATCH FOR

Treating independence and disjointness as similar because both sound like 'unrelated'. Disjoint events are maximally DEPENDENT: knowing one occurred tells you the other definitely did not.

CHECK YOURSELF · ANSWERS AT THE BACK

21.1 A and B are disjoint with $P(A), P(B) > 0$. Are they independent?

A Yes — disjoint events never interact
B No — $P(A \cap B) = 0$ while $P(A)P(B) > 0$, so they are strongly dependent
C Only if $P(A) = P(B)$
D It cannot be determined

21.2 The probability of at least one six in six fair dice, to three decimal places, is:

Your answer: _____

21.3 The probability of at least three sixes in eighteen fair dice, to three decimal places, is:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 22 OF 22

The clues to a great story — make me care

Sun 18 Oct · 19 min video · Watch on YouTube

THE TAKEAWAY

Stanton's craft notes from Pixar: know the punchline, make a promise, make the audience care, and let them assemble the meaning themselves — two plus two, never four.

THE ARGUMENT · 7 STEPS

1. **Storytelling is joke telling**
   Know your ending, and make everything from the first sentence to the last lead toward it.

2. **Make me care**
   The greatest story commandment. You stop flicking channels on the one programme that made you care, and that is by design.

3. **The promise**
   A good beginning promises the story will be worth the time — like a pebble pulled back in a slingshot.

4 **Two plus two**
Audiences are born problem-solvers and want to work for their meal without noticing. A well-organised absence of information draws them in.

5 **Anticipation mingled with uncertainty**
Short-term tension (will Dory forget?) running inside long-term tension (will they find Nemo?), built on honest conflict.

6 **Spine, change, theme, wonder**
Characters have an inner drive they cannot escape; stories die when things go static; a theme runs underneath; and the rarest ingredient is wonder.

7 **Use what you know**
Not plot or fact, but a truth you have felt — Marlin's promise to Nemo came from a boy told he would not live.

WORKED EXAMPLE

Take the emotional peak of your story and remove the sentence that names the emotion. Replace it with two concrete details from which the listener can work the emotion out. That is two plus two.

WATCH FOR

Giving the audience four: explaining the feeling, the lesson and the meaning outright. It feels clear to the teller and leaves the listener with nothing to do — and nothing to care about.

CHECK YOURSELF · ANSWERS AT THE BACK

22.1 What does Stanton mean by 'give them two plus two, not four'?

A Give the audience two examples for every point you want them to learn
B Let the audience assemble the meaning from the clues you provide
C Tell the audience the meaning twice, so nobody can possibly miss it
D Keep each story to two short scenes, so it ends in under four minutes

22.2 How does he define drama, quoting William Archer?

A Conflict between what we want and fear
B A beginning, a middle and a turning point
C Anticipation mingled with uncertainty
D Surprise followed quickly by relief

22.3 What does he call the greatest story commandment?

A Make me care
B Know your audience
C Never use dialogue
D End with a twist

WEEK 2 · MON 12 – SUN 18 OCTOBER 2026

What this week established

Each lecture’s takeaway, by course.

MATH 110 · Linear Algebra

1 **Cramer's rule, explained geometrically.** Each coordinate of the unknown vector is an area, and every area scales by the determinant — that is the whole of Cramer's rule.

2 **Change of basis.** A matrix whose columns are someone else's basis vectors translates their coordinates into yours; its inverse goes the other way.

3 **Eigenvectors and eigenvalues.** An eigenvector is a direction the transformation only stretches, never turns, and its eigenvalue is that stretch.

4 **A quick trick for computing eigenvalues.** The trace and the determinant hand you the mean and the product of the eigenvalues, and two numbers are pinned down by those.

14 **Abstract vector spaces.** A vector is anything you can sensibly add and scale, and every result in linear algebra applies to all of them at once.

15 **The Geometry of Linear Equations.** Every linear system has a row picture and a column picture, and the column picture is the one that survives into higher dimensions.

AI 200 · Neural Networks: Zero to Hero

5 **Makemore 1 — bigram character-level language model.** A bigram language model can be built two ways — count which character follows which and normalise, or train a one-layer net with a softmax on the negative log likelihood — and the two arrive at the same model, which is the point: the neural framing is the one that scales.

13 **Makemore 2 — MLP, following Bengio et al. 2003.** To see more than one character back, stop counting and learn: embed each character as a small vector, concatenate the last three, and feed them through a hidden layer — Bengio et al.'s MLP — trained on minibatches and judged on data it never trained on.

20 **Makemore 3 — activations, gradients, BatchNorm.** A deep net trains well only if its activations and gradients stay well scaled from the first step: start with an output that is unconfident, hidden units that are not saturated, weights scaled by fan-in — and batch normalisation to hold every layer's pre-activations near unit Gaussian — then watch the statistics instead of guessing.

SPCH 100 · Storytelling

6 **What is a story — and why tell one.** Dicks's case is that a told story is remembered, earns trust and changes the teller — and that telling one well is a matter of deciding what to say before you say it, not of having lived a dramatic life.

11 **The five-second moment: the instant a story is about.** A story is about the instant someone changes — usually small and internal — and the dramatic events around it only matter as the road to that instant.

16 **Anecdote, bait and reflection — the two building blocks.** Ira Glass reduces a story to two building blocks — the anecdote, which carries momentum, and the moment of reflection, which carries meaning — and says the craft is weaving them together while baiting the listener with questions.

17 **The elephant: a reason to listen in the first thirty seconds.** Dicks's essential stake is the elephant: an early, clear signal of what the story is about and what could go wrong, which gives the audience a reason to stay — and which may turn into a different stake as the story develops.

22 **The clues to a great story — make me care.** Stanton's craft notes from Pixar: know the punchline, make a promise, make the audience care, and let them assemble the meaning themselves — two plus two, never four.

MATH 120 · Single-Variable Calculus

7 **Limits, L'Hôpital's rule, and epsilon delta definitions.** Limits exist to avoid talking about infinitely small changes, and once you have them the derivative can turn around and evaluate limits.

8 **Integration and the fundamental theorem of calculus.** The integral is the limit of a sum of thin rectangles, and you evaluate it by finding an antiderivative and looking at just two inputs.

9 **What does area have to do with slope?** Averaging a continuous quantity is an integral divided by a width — and reading that as an average slope explains why antiderivatives solve integrals at all.

10 **Higher order derivatives.** The second derivative is the derivative of the derivative — visually it is curvature, and physically it is acceleration.

18 **Taylor series.** Taylor series translate derivative information at a single point into approximation information near that point, by matching one derivative per coefficient.

19 **The other way to visualize derivatives.** Read a function as a transformation of the number line and the derivative becomes a local stretch factor — a view that survives into contexts where graphs do not.

MATH 130 · Probability

12 **Birthday Problem, Properties of Probability.** Twenty-three people suffice for a better-than-even birthday match, because the quantity that matters is not k but the $\binom{k}{2}$ PAIRS — and inclusion-exclusion is the general tool for the probability of a union.

21 **Conditional Probability.** Conditional probability is $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$ — restrict the world to B and renormalise — and conditioning is the soul of statistics, because it is how beliefs get updated by evidence.

WEEK 2 · MON 12 – SUN 18 OCTOBER 2026

Answers

The answers and explanations are the summaries' own.

1 Cramer's rule, explained geometrically

1.1 B — Scales every area by the same factor, its determinant

One common factor for every area is what lets you divide it back out. Preserving areas would make the determinant 1, which is a special case, not the rule.

1.2 2

$\det \begin{pmatrix} 9 & 1 \\ 8 & 2 \end{pmatrix} = 18 - 8 = 10$, divided by $\det A = 6 - 1 = 5$.

2 Change of basis

2.1 B — The same transformation M, described in the other basis

Translate in, transform, translate back. The transformation is unchanged; only the language describing it has moved.

2.2 −4

$-1 \cdot 2 + 2 \cdot (-1) = -2 - 2 = -4$.

3 Eigenvectors and eigenvalues

3.1 B — A nonzero vector can only be sent to zero if the map collapses space

$(A - \lambda I)v = 0$ with $v \neq 0$ requires a nontrivial null space, which means zero determinant.

3.2 3

Triangular, so the eigenvalues are the diagonal entries 3 and 2.

4 A quick trick for computing eigenvalues

4.1 B — The determinant

p is the product of the eigenvalues, and the product of the eigenvalues is the determinant. m is the mean, which comes from the trace.

4.2 9

Mean of the diagonal is 5; determinant is $16 - 7 = 9$. So $\lambda = 5 \pm \sqrt{25 - 9} = 5 \pm 4$, giving 9 and 1.

5 Makemore 1 — bigram character-level language model

5.1 B — A start only ever opens a bigram and an end only closes one, so they never clash

A start token only ever appears as the first character of a bigram and an end token only as the second, so one symbol can play both roles without ambiguity. With two tokens the table had a row for the end token and a column for the start token that could only ever hold zeros.

5.2 9

'.isabella.' has 10 symbols, so 9 adjacent pairs: one for the first letter, seven between letters, and one for the end. In general a name of n letters gives n + 1 examples — 'emma' gives the 5 shown in the notebook.

5.3 D — Without keepdims the sums broadcast as a row, and columns get divided

P.sum(1, keepdims=True) is 27 × 1, which stretches across the columns, so row i is divided by row i's total. P.sum(1) is shape 27; broadcasting aligns it from the right as 1 × 27, so column j is divided by row j's total — silently wrong.

6 What is a story — and why tell one

6.1 B — Deciding what to say before saying it

He calls storytelling decision-making: simple strategies anyone can learn. He says he is not gifted, just someone who thinks before he speaks.

6.2 C — So the audience likes and trusts him before he asks them to act

He says it outright: a story earns trust and makes people want to follow you — persuasion, but benevolent.

6.3 A — A habit from a hungry childhood still ran his behaviour

Children who grow up hungry learn never to accept food they cannot reciprocate. He would not have known without asking.

7 Limits, L'Hôpital's rule, and epsilon delta definitions

7.1 B — Some ε exists for which no δ is small enough to trap the outputs

Being undefined at the point is normal — that is exactly the case limits handle. Failure means you can name a tolerance, say 0.4, that the output range never fits inside no matter how tight you squeeze the inputs.

7.2 4

Both parts vanish at $x = 2$. Differentiating separately gives $2x$ over 1, and at $x = 2$ that is 4. Factoring as $(x + 2)(x - 2)/(x - 2)$ gives the same answer.

8 Integration and the fundamental theorem of calculus

8.1 B — Nudging the right edge by dT adds a sliver that is nearly a rectangle of height $f(T)$ and width dT

$dA \approx f(T)\,dT$, so $dA/dT = f(T)$. The approximation improves as dT shrinks, which makes the limit exact — and that is the fundamental theorem.

8.2 72

The antiderivative is $4t^2 - \frac{1}{3}t^3$. At $t = 6$ that is $144 - 72 = 72$; at $t = 0$ it is 0. So the integral is $72 - 0 = 72$.

9 What does area have to do with slope?

9.1 B — The slope of the antiderivative's graph between the two endpoints

Rise over run on F. Since f is by definition the slope of F at each point, averaging f is averaging those tangent slopes — and the average slope over a stretch is the overall slope end to end.

9.2 2

The antiderivative is $-\cos x$, which is $-(-1) = 1$ at π and -1 at 0. The change is $1 - (-1) = 2$.

10 Higher order derivatives

10.1 B — A negative second derivative, because the slope is decreasing there

Curvature direction IS the sign of the second derivative. The first derivative can be anything — steeply positive, zero, negative — while the graph bends downwards.

10.2 18

$f'(x) = 3x^2$ and $f''(x) = 6x$, so at $x = 3$ it is 18. The graph of x^3 is bending upwards there, and the positive sign agrees.

11 The five-second moment: the instant a story is about

11.1 C — Realising, as his friends crowd the doorway, that he is loved

The crash is the road. The change is from feeling alone to knowing he has a family.

11.2 A — Nobody can connect to the experience of going through a windshield

People wince at gore but cannot connect to it. The emotional response comes at the moment they recognise from their own lives.

11.3 D — That playing golf with his son kept a promise he'd made as a boy

He let the moment roll around for three months; the realisation arrived later, which is why he argues for being curious about yourself.

12 Birthday Problem, Properties of Probability

12.1 B — Because there are $\binom{23}{2} = 253$ pairs, and 253 is comparable to 365

The event is about pairs, not people, and pairs grow quadratically. That is why the intuitive answer — something over 100 — is so far out.

12.2 0.507

$1 - \frac{365 \cdot 364 \cdot \cdots \cdot 343}{365^{23}} = 0.5073$. The first k for which it exceeds a half.

12.3 0.632

$1 - \frac{1}{2!} + \frac{1}{3!} - \cdots \to 1 - e^{-1} = 0.6321$. Equivalently $1 - D_n/n!$, where D_n counts the derangements.

13 Makemore 2 — MLP, following Bengio et al. 2003

13.1 11897

Embedding 27 × 10 = 270; hidden layer 30 × 200 + 200 = 6,200; output layer 200 × 27 + 27 = 5,427. Total 11,897 — the notebook prints the same. Note the embedding table counts: it is learned like any other weight.

13.2 5

Every character of 'emma.' is a target — e, m, m, a and the final '.' — each predicted from the three before it, padded with '.': '...' predicts e, '..e' predicts m, and so on. n letters give n + 1 examples.

13.3 B — To lay the three 10-number embeddings side by side as one input

C[X] has shape (batch, 3, 10). The hidden layer wants one vector per example, so the three embeddings are concatenated into 30 numbers — keeping which character was where, which averaging would lose. The -1 lets PyTorch infer the batch size.

14 Abstract vector spaces

14.1 B — Let a result proved once apply to every set that satisfies them

They are a checklist someone else verifies about their own objects. Prove a theorem from the axioms and it reaches every space anyone will ever define.

14.2 10

The derivative is $3x^2 + 10x + 4$; the x term comes from differentiating $5x^2$.

15 The Geometry of Linear Equations

15.1 B — Combinations of the columns fill the whole space

The reachable right-hand sides are precisely the combinations of the columns. If those fill the space, every $\vec{b}$ is reachable; if they fill only a plane, most are not.

15.2 3

$1 \cdot (-1) + 2 \cdot 2 = -1 + 4 = 3$.

16 Anecdote, bait and reflection — the two building blocks

16.1 B — The anecdote and the moment of reflection

The anecdote is the sequence of actions; the reflection says why it matters.

16.2 A — The unusual quiet, which raises the question why

The quiet plants a question, and any question you raise implies you will answer it.

16.3 C — A thrilling anecdote that means absolutely nothing

Without a moment of reflection, even a surprising sequence of events tells the listener nothing new.

17 The elephant: a reason to listen in the first thirty seconds

17.1 B — Films have trailers, so their audience already knows the stakes

A film audience arrives knowing what kind of story it is. A listener has no idea until you tell them.

17.2 D — The main stake can shift as the story turns, as on the widower's porch

The first stake gets the audience in; a new one can take over once the story turns.

17.3 A — My tyre has just disintegrated, I have no phone and no money, and I am a hundred miles from home.

It gives the listener a question to hold — how will he get home? — without summarising the story.

18 Taylor series

18.1 B — Differentiating x^n n times produces a factor of $n!$, which must be cancelled

Each application of the power rule drops the exponent in front, so n of them leave $1 \cdot 2 \cdots n$. Dividing by $n!$ makes the n-th derivative of the polynomial come out equal to $f^{(n)}(0)$ exactly.

18.2 0.995

$1 - \frac{0.01}{2} = 1 - 0.005 = 0.995$, which matches the true value to the digits shown. The next correction is $\frac{0.0001}{24}$, about four millionths.

19 The other way to visualize derivatives

19.1 B — $|f'|$ there is less than 1, so nearby points contract towards it

Each pass multiplies the distance from the fixed point by roughly $|f'|$. Below 1 that distance shrinks to nothing; above 1 it grows, and the point repels whatever you start with.

19.2 6

The stretch factor is the derivative: $\frac{d}{dx}x^2 = 2x$, which is 6 at $x = 3$. Near $x = 1$ the same function stretches by only 2.

20 Makemore 3 — activations, gradients, BatchNorm

20.1 D — So the first predictions are near uniform, not confidently wrong

With near-zero logits every character gets probability about 1/27, so the first loss is about $\ln 27 \approx 3.30$ — the notebook's first step prints 3.32. Random large logits would be confidently wrong, the loss would start far higher, and the first steps would be spent just squashing them. The loss log calls this fix 'softmax confidently wrong'.

20.2 0.304 (± 0.001)

fan_in = 30, $\sqrt{30} \approx 5.477$, and $(5/3)/5.477 \approx 0.304$. Dividing by $\sqrt{\text{fan_in}}$ keeps the pre-activations' std near 1 however many inputs a unit sums; 5/3 is the gain that compensates for tanh squashing.

20.3 B — There tanh's local gradient, 1 – t², is near zero, so little flows back

Backprop through tanh multiplies the incoming gradient by $1 - t^2$. At $|t|$ = 0.97 that factor is about 0.06; at ±1 it is 0. A unit that is saturated for every example never learns — a dead neuron. The loss log's 'tanh layer too saturated at init' fix scales W1 down for exactly this.

21 Conditional Probability

21.1 B — No — $P(A \cap B) = 0$ while $P(A)P(B) > 0$, so they are strongly dependent

Knowing A occurred tells you B certainly did not, which is the most informative thing one event can say about another.

21.2 0.665

$1 - (5/6)^6 = 0.6651$. This is Pepys's option A, and the largest of the three — against his own strong belief.

21.3 0.597

$1 - \sum_{k=0}^{2} \binom{18}{k}(1/6)^k(5/6)^{18-k} = 0.5973$ — the least likely of the three, and the one Pepys bet on.

22 The clues to a great story — make me care

22.1 B — Let the audience assemble the meaning from the clues you provide

The audience wants to work for its meal; the order and choice of elements decides whether they engage.

22.2 C — Anticipation mingled with uncertainty

Have you made the audience want to know what happens next, and how it will all end?

22.3 A — Make me care

Emotionally, intellectually or aesthetically — just make me care.